

Template for Arduino and Motor Activity

Statement Given:

Task 1: Interfacing of Servo Motor to Arduino

- a) Make the servo rotate back and forth (0-180 degrees) using the Arduino
- b) Running Servo Motor using TinkerCAD

Task 2 : DC Motor Speed and Direction Control

- a) Control Positioning of Servo Motor changing from 20 to 180 degrees and Vice Versa, try changing values
- b) Try Running DC Motor in Forward and Reverse directions and control the Speed of the DC Motor **using TinkerCAD**

Evaluation Criteria:

1. Connections as per task given,
2. Code for Arduino for the Task.
3. Successful execution of the activities.

Performance - 4 Marks, Servo using TinkerCAD - 4 marks , DC Motor using TinkerCAD - 4 marks, Submission - 3 Marks

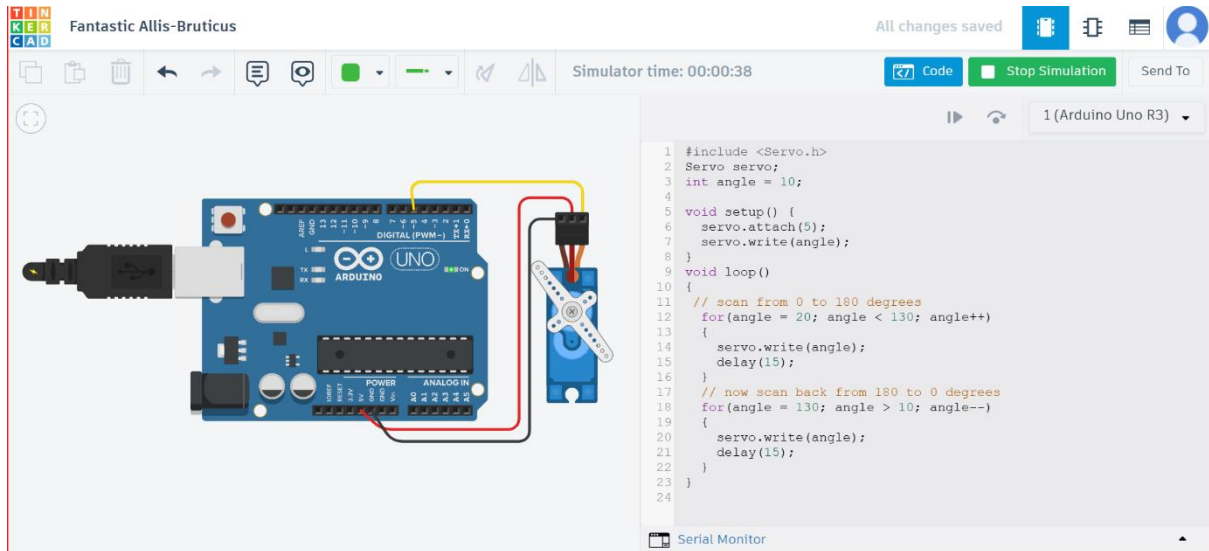
Team

Sr. No.	Roll No	Name	Work Done
1	16014325030	Avaneesh majumder	Connections for motors
2	16014325031	Vijayta malkani	coding
3	16014325032	Sarbani mandal	Tinker cad simulation
4	16014325033	Om mane	coding
5	16014325034	Aarav mody	Tinker cad simulation

Fill in your details as per the following points

- **Code for the task given**
- **Photo and Video of Actual Implementation**
- **Observations, reflection on activity**

Task 1: Interfacing of Servo Motor to Arduino



Code for Servo Motor

```
#include <Servo.h>
Servo servo;
int angle = 10;

void setup() {
  servo.attach(5);
  servo.write(angle);
}

void loop()
{
  // scan from 0 to 180 degrees
  for(angle = 20; angle < 130; angle++)
  {
    servo.write(angle);
    delay(15);
  }
  // now scan back from 180 to 0 degrees
  for(angle = 130; angle > 10; angle--)
  {
    servo.write(angle);
    delay(15);
  }
}
```

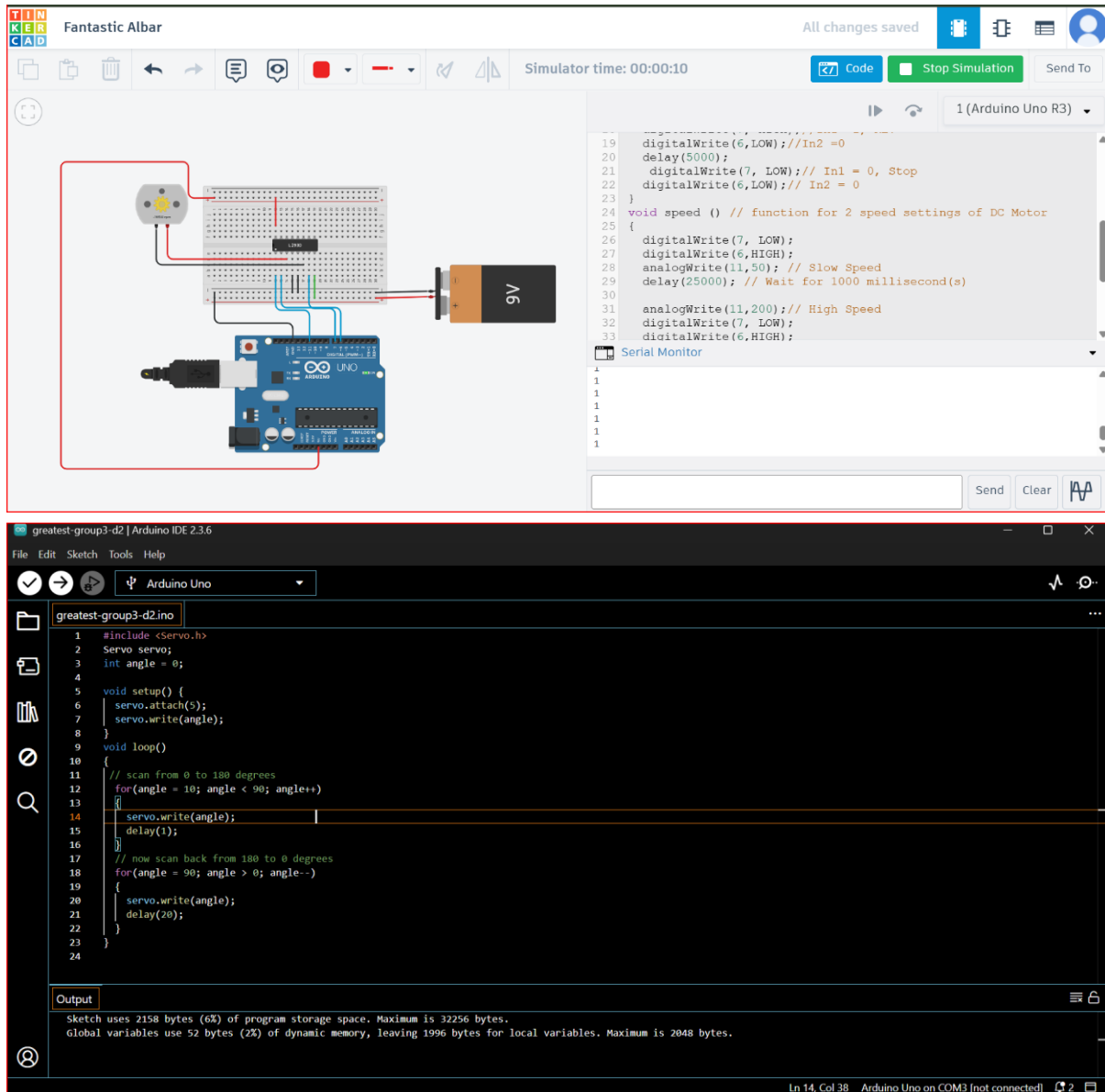
Observation

The servo motor responded accurately to the control signals given through the Arduino Uno. In the Tinkercad simulation, the motor rotated smoothly to specific angles based on the program. When implemented manually using Arduino Uno, the servo motor showed similar behavior.

Reflection

This activity helped in understanding how servo motors work using PWM signals and how Arduino controls precise angular movement.

Task 2 : DC Motor Speed and Direction Control



The image displays two screenshots from a TINKERCAD project. The top screenshot shows a circuit diagram in the TINKERCAD simulator. An Arduino Uno R3 is connected to a breadboard. A 9V battery is connected to the breadboard. A DC motor is connected to the breadboard. The code in the simulator is as follows:

```
19 digitalWrite(6, LOW); // In2 = 0
20 delay(5000);
21 digitalWrite(7, LOW); // In1 = 0, Stop
22 digitalWrite(6, LOW); // In2 = 0
23 }
24 void speed () // function for 2 speed settings of DC Motor
25 {
26   digitalWrite(7, LOW);
27   digitalWrite(6, HIGH);
28   analogWrite(11, 50); // Slow Speed
29   delay(25000); // Wait for 1000 millisecond(s)
30 }
31 analogWrite(11, 200); // High Speed
32 digitalWrite(7, LOW);
33 digitalWrite(6, HIGH);
```

The bottom screenshot shows the Arduino IDE 2.3.6 interface. The sketch is named "greatest-group3-d2.ino". The code is as follows:

```
1 #include <Servo.h>
2 Servo servo;
3 int angle = 0;
4
5 void setup() {
6   servo.attach(5);
7   servo.write(angle);
8 }
9 void loop() {
10   // scan from 0 to 180 degrees
11   for(angle = 10; angle < 90; angle++)
12   {
13     servo.write(angle);
14     delay(1);
15   }
16   // now scan back from 180 to 0 degrees
17   for(angle = 90; angle > 0; angle--)
18   {
19     servo.write(angle);
20     delay(20);
21   }
22 }
23
24
```

The output window shows the following message:

```
Sketch uses 2158 bytes (6%) of program storage space. Maximum is 32256 bytes.
Global variables use 52 bytes (2%) of dynamic memory, leaving 1996 bytes for local variables. Maximum is 2048 bytes.
```

Code for motors

```
void setup()
{
  pinMode(11, OUTPUT); // En of Motor 1
  pinMode(7, OUTPUT); // In1 of Motor 1
  pinMode(6, OUTPUT); // In 2 of Motor 1
}

void fwdrev() // function reverses direction of motor
{
  digitalWrite(7, LOW); // In1 = 0, FWD
  digitalWrite(6, HIGH); // In2 = 1
}
```

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```
digitalWrite(11, HIGH);

delay(5000); // Wait for 5 second(s)
digitalWrite(7, LOW); // In1 = 0, Stop
digitalWrite(6, LOW); // In2 = 0
delay(5000); // Wait for 1000 millisecond(s)
digitalWrite(7, HIGH); // In1 = 1, REV
digitalWrite(6, LOW); // In2 = 0
delay(5000);
digitalWrite(7, LOW); // In1 = 0, Stop
digitalWrite(6, LOW); // In2 = 0
}
void speed () // function for 2 speed settings of DC Motor
{
digitalWrite(7, LOW);
digitalWrite(6, HIGH);
analogWrite(11, 50); // Slow Speed
delay(25000); // Wait for 1000 millisecond(s)

analogWrite(11, 200); // High Speed
digitalWrite(7, LOW);
digitalWrite(6, HIGH);
delay(25000); // Wait for 1000 millisecond(s)
}

void loop()
{
fwdrev();
speed();
}
```

Observation

In the Tinkercad simulation, the DC motor rotated continuously when powered by a 9V battery. The RPM value was displayed and changed based on the applied voltage and circuit conditions, indicating variation in motor speed.

Reflection

This activity helped in understanding the basic working principle of DC motors and the importance of proper power sources.

GOOGLE DRIVE LINK FOR VIDEOS

https://drive.google.com/drive/folders/1GM84GMMpW-5ZB8a37SrWZ9mWBq_IW9b5?usp=sharing